MARKING SCHEME

General: Accept any correct biological wording. Do not deduct for spelling unless the meaning becomes unclear.

Question 1 [3 marks]
- A first valid difference stated as a contrast (e.g. no membrane-bound nucleus vs nucleus; no membrane-bound organelles vs present; 70S vs 80S ribosomes; circular vs linear DNA; smaller vs larger size). (1 mark)
- A second, different valid difference. (1 mark)
- A third, different valid difference. (1 mark)

Question 2 [4 marks]
- Mitochondria are the site of aerobic respiration (Krebs cycle / electron transport chain). (1 mark)
- They produce ATP, the cell's usable energy. (1 mark)
- Muscle cells need a lot of energy (ATP) for contraction. (1 mark)
- Links the two: more mitochondria give more ATP production to meet that demand. (1 mark)

Question 3 [8 marks] - mark holistically using these bands:
7-8 marks: Explains the active site and substrate specificity (lock-and-key or induced fit), formation of the enzyme-substrate complex and lowering of activation energy; describes the effect of temperature up to an optimum (kinetic energy, collisions) and denaturation above it (active site changes shape); describes optimum pH and how pH changes alter bonds in the active site. Correct terminology throughout.
5-6 marks: Most of the points above, with minor gaps or imprecise terms.
3-4 marks: Basic description: mentions the active site and that temperature and pH have an optimum, but little explanation.
1-2 marks: A few relevant points only, or significant misconceptions (e.g. enzymes are "killed").
0 marks: No relevant content.